Right Once, Wrong Next Time: Repeated-Trial Reliability of Small Local Models on Roman Urdu Clinic Tool Calls

Muzammal Hussain
Independent researcher, Glasgow, United Kingdom
muzammal@masmatixinnovations.co.uk
ORCID: 0009-0001-2570-5686

November 2026

Abstract

Tool-use evaluations usually score one attempt per task, but an agent that books the right appointment four times out of five is not safe to leave unattended. We measure pass^k, the probability that all k attempts at the same task are correct, for two small open-weight models run locally (`llama3.1:8b` and `qwen2.5:3b`) on eight physiotherapy-clinic tool calls requested in English and in Roman Urdu, the Latin-script Urdu that many people in Pakistan and its diaspora type. Each task was attempted five times in two ways: with five wordings of the same request at temperature 0, and with one wording sampled five times at temperature 0.7 (320 calls). In English, pass^5 was 87.5% for `llama3.1:8b` and 75.0% for `qwen2.5:3b` in both conditions. In Roman Urdu it fell to 12.5–37.5%, and single attempts hid the gap: with five wordings, `qwen2.5:3b` got 62.5% of single attempts right but only 12.5% of tasks right every time. Rewording made Roman Urdu tasks intermittently wrong; sampling made them systematically wrong, with up to half of the tasks failing in all five samples, so a retry would not have helped. The failures concentrated in values that need interpretation rather than copying: times written as words (*dhai baje*, half past two), number words, a refund amount, a cancellation reason and a city. 67 of the 73 failed calls (91.8%) were valid against the tool schema, and in an earlier pilot a validator-feedback loop never triggered while a quarter of calls carried wrong values. For clinic agents in code-mixed languages, schema validation is not a safety check; repeated-trial evaluation and read-back confirmation of interpreted values are needed. All prompts, replies and analysis code are public.

Keywords: LLM agents; tool calling; pass^k; reliability; Roman Urdu; code-mixed language; local language models; clinical software

1 Introduction

An agent acts through tool calls: it books an appointment, records a pulse or refunds a payment by filling a structured argument list. Benchmarks usually report how often the first attempt is correct. For deployment the more useful question is whether the agent is correct *every* time it meets the same request, because a clinic runs the same kinds of request hundreds of times a week. τ-bench introduced pass^k, the probability that all k independent attempts at a task succeed, and found that even state-of-the-art function-calling agents succeeded on fewer than half of the tasks and that pass^8 fell below 25% in the retail domain [1]. That study used English and large hosted models, and its repeated attempts varied both the simulated user's wording and the model's sampling.

Small clinics, especially in Pakistan and among its diaspora, have two further constraints. Patient data should stay on local hardware, which favours models of a few billion parameters. And staff often write in Roman Urdu: Urdu in the Latin alphabet with English terms mixed in and no fixed spelling, a setting in which language technology is scarce [2, 3] and code-mixed text is known to be hard for NLP systems [4]. In earlier work we found that Roman Urdu input reduced the accuracy of small local ReAct agents on clinic questions by up to 35 points [5]. Here we ask the narrower question of single tool calls:

Table 1: Examples of the requests and the values the tool call must contain.

Tool	Roman Urdu request	Required values
Book appointment	*Patient number 4417 ki appointment 3 November 2026 ko dopahar dhai baje book karo. Urgent nahi hai.*	time 14:30, urgent false
Record vitals	*Patient 2093 ke vitals save karo: BP 128/84, pulse bahattar.*	128, 84, pulse 72
Refund	*Invoice INV-1029 ke paintalees pound pachaas pence wapas kar do.*	amount 45.50, GBP
Cancel	*Appointment 55821 cancel karo kyunke physio beemar hai, aur patient ko bata do.*	reason clinician_unavailable, notify true
Search	*Glasgow mein Akhtar surname wale zyada se zyada paanch patients dhoondo.*	last name Akhtar, city Glasgow, 5

1. How repeatable are small local models' clinic tool calls in English and in Roman Urdu, measured by pass^k?
2. Is the variation driven by how the request is worded or by random sampling?
3. Which arguments fail, and would a schema validator catch them?

2 Method

Tasks. Eight clinic tools, each with a JSON schema and a ground-truth argument list: book an appointment, record vital signs, record a pain score, set a reminder, refund a payment, cancel an appointment, search patients, and log an exercise. Each request was written in English and in Roman Urdu (Table 1) and deliberately includes values given in words: "half past two in the afternoon"/*dopahar dhai baje*, "seventy-two"/*bahattar*, "forty-five pounds fifty"/*paintalees pound pachaas pence*.

Two sources of variation. Each task was attempted five times per language in two conditions. *Wording*: the original request and four rewordings with the same meaning, at temperature 0 and seed 42; the Roman Urdu rewordings use the spelling variety people actually type (*kr/kar*, *dopehar/dopahar*, *nai/nahi*). *Sampling*: the original wording at temperature 0.7 with seeds 1–5. The grid is 2 models $\times$ 8 tasks $\times$ 2 languages $\times$ 2 conditions $\times$ 5 attempts = 320 calls.

Models and scoring. `qwen2.5:3b` [6] and `llama3.1:8b` [7] ran through Ollama on a MacBook (Apple M4, 16 GB) with at most 256 output tokens; a call that timed out after 120 s was retried once (none did in the final run). An attempt passes if the reply is valid against the tool schema and every value matches the ground truth after normalising case and punctuation. We report pass^k with the unbiased estimator of τ-bench, $\binom{c}{k}/\binom{n}{k}$ averaged over tasks, with $n = 5$ attempts and c passes, and 95% percentile bootstrap intervals over the eight tasks (2,000 resamples) [8]. We also count tasks that always, sometimes or never pass, and the number of distinct outputs per task.

Validator pilot. Before this study we ran each of the eight tasks once per language with `qwen2.5:3b` at temperature 0 inside a self-correction loop: if a schema validator found problems, they were sent back to the model for up to two retries, without the expected values.

Table 2: pass[1] and pass[5] (%) with 95% bootstrap intervals over the 8 tasks, task outcome counts and the mean number of distinct outputs per task.

Model	Language	Condition	pass[1]	pass[5]	Always / some / never	Distinct
`llama3.1:8b`	English	wording	97.5 [92.5, 100]	87.5 [62.5, 100]	7 / 1 / 0	1.12
`llama3.1:8b`	English	sampling	97.5 [92.5, 100]	87.5 [62.5, 100]	7 / 1 / 0	1.12
`llama3.1:8b`	Roman Urdu	wording	80.0 [67.5, 92.5]	37.5 [0.0, 75.0]	3 / 5 / 0	1.88
`llama3.1:8b`	Roman Urdu	sampling	47.5 [12.5, 82.5]	37.5 [12.5, 75.0]	3 / 1 / 4	1.62
`qwen2.5:3b`	English	wording	95.0 [87.5, 100]	75.0 [37.5, 100]	6 / 2 / 0	1.25
`qwen2.5:3b`	English	sampling	85.0 [60.0, 100]	75.0 [37.5, 100]	6 / 1 / 1	1.12
`qwen2.5:3b`	Roman Urdu	wording	62.5 [47.5, 77.5]	12.5 [0.0, 37.5]	1 / 7 / 0	2.50
`qwen2.5:3b`	Roman Urdu	sampling	52.5 [22.5, 80.0]	25.0 [0.0, 62.5]	2 / 3 / 3	1.62

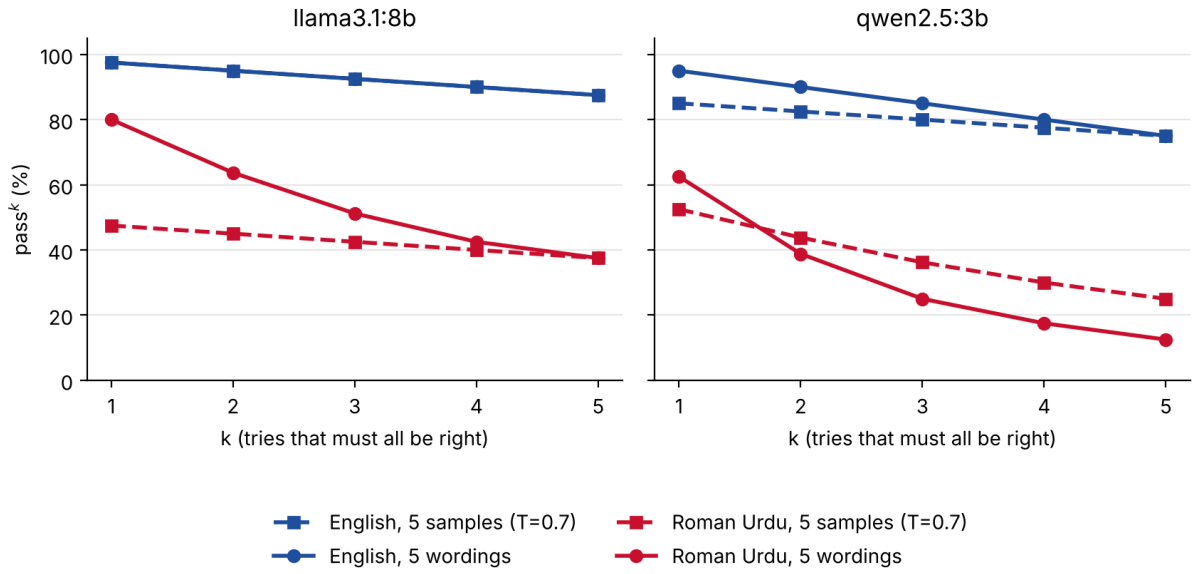


Figure 1: pass^k for $k = 1$ to 5. Blue: English; red: Roman Urdu. Solid: five wordings at temperature 0; dashed: one wording sampled five times at temperature 0.7.

3 Results

3.1 English is repeatable, Roman Urdu is not

Table 2 and Figure 1 give pass^k for $k = 1$ to 5. In English, both models were mostly repeatable: pass[5] was 87.5% for `llama3.1:8b` (7 of 8 tasks right in every attempt) and 75.0% for `qwen2.5:3b`, identical across the two conditions. In Roman Urdu pass[5] fell to 37.5% for `llama3.1:8b` and to 12.5% (wording) and 25.0% (sampling) for `qwen2.5:3b`. Single attempts understated the problem most when the wording varied: `llama3.1:8b` went from 80.0% at $k = 1$ to 37.5% at $k = 5$, and `qwen2.5:3b` from 62.5% to 12.5%. With eight tasks the intervals are wide; the gap between languages is clear, but differences between models and conditions are not.

3.2 Wording and sampling fail differently

The two conditions produced different kinds of failure in Roman Urdu. Rewording made tasks *intermittently* wrong: for `llama3.1:8b` 5 of 8 tasks and for `qwen2.5:3b` 7 of 8 were right in some wordings and wrong in others, and `qwen2.5:3b` produced 2.5 distinct outputs per task on average. Sampling the original wording made failures *systematic*: `llama3.1:8b` got 4 of the 8 Roman Urdu tasks wrong in all five samples (the appointment time, the pulse, the refund amount and the cancellation reason), and `qwen2.5:3b` got 3 wrong in all five. For those tasks, retrying with a new random seed would not have

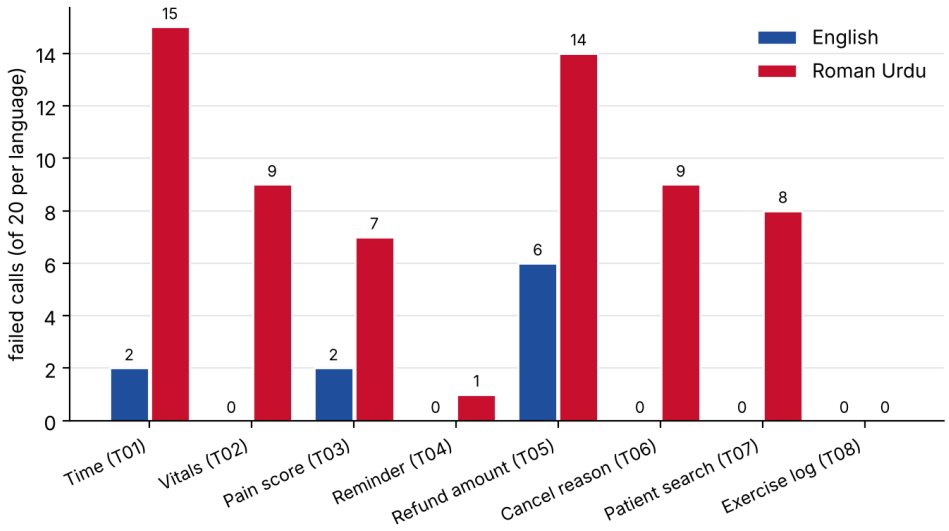


Figure 2: Failed calls per task and language, out of 20 per task and language (2 models × 2 conditions × 5 attempts).

helped; the model consistently read the request the wrong way.

3.3 Which values break

Of the 320 calls, 73 failed: 63 in Roman Urdu and 10 in English (Figure 2). The failures were in values that require interpretation rather than copying. The time *dopahar dhai baje* (14:30) was written as 14:00 (7 calls) or 13:00 (6), so "*dhai*" (two and a half) lost its half. The refund of 45.50 became 45.0 (6 calls), and in 4 calls 100, a number that does not appear in the request. The pulse *bahattar* (72) became 100 in 4 calls. "Physio *beemar* hai" (the physiotherapist is ill) was mapped to the generic reason `other` instead of `clinician_unavailable` in 6 calls. The city in "*Glasgow mein*" (in Glasgow) was left empty in 7 calls. The exercise log, where every value is a plain number or a word copied from the request, was right in all 40 attempts in both languages. In English the errors were mostly the same refund fraction (6 calls dropped the 50 pence).

3.4 A schema validator would not have caught them

67 of the 73 failed calls (91.8%) were valid against the tool schema: correct keys, types and formats, with a wrong value. An empty city is still a string; 14:00 is a valid time. The earlier validator pilot shows the same from the other side: all 16 calls of `qwen2.5:3b` passed the validator at the first attempt, so the self-correction loop never ran, yet 4 of them (3 in Roman Urdu) carried a wrong value, including the invented refund of 100. Output formatting was a minor issue by comparison: 150 of the 320 replies wrapped the JSON in code fences despite being told not to, but they could still be parsed.

4 Discussion

Score more than one attempt. For English the single-attempt score was a fair guide to repeatability. For Roman Urdu it was not: a model that is right on 80% of single attempts can be right on every attempt for only 37.5% of tasks. Evaluations of agents for code-mixed or low-resource languages should report pass^k or the number of tasks that are always right, not only accuracy.

Vary the wording, not only the seed. Sampling and rewording expose different risks. Rewording shows that the same meaning in a different spelling or phrasing can flip a call; sampling shows that some

requests are read wrongly every time. A test set built from one wording per task and repeated seeds would miss the first kind; one attempt per wording would miss the second.

Validate meaning, not only shape. Almost all failures were schema-valid. Stricter schemas help with some (no empty strings, a plausible range for a pulse), but an appointment at 14:00 instead of 14:30 is only caught by comparing the call with what the user meant. For a clinic tool, a read-back step ("Book 4417 on 3 November at 14:30, is that right?") is the simplest safeguard for interpreted values such as times, amounts and doses.

Number words are the weak point. The values that broke most were numbers and times written as words in Roman Urdu, together with the free-text reason and city. Normalising number words to digits before the model sees them is a cheap intervention that this study suggests but does not test.

5 Limitations

Eight tasks is a pilot: one task is 12.5 percentage points, and the bootstrap intervals are wide. The requests and rewordings were written by the author and are not a sample of real clinic messages; other Roman Urdu spellings may behave differently. Each call is a single turn, whereas τ-bench scores full conversations. Scoring is exact match after light normalisation, so a correct paraphrase of a free-text field counts as wrong. Ollama seeds make runs repeatable on one machine but not necessarily across versions or hardware. The validator pilot used one model and 16 calls. No clinician reviewed the tasks, and no patient data were used.

6 Conclusion

Two small local models made clinic tool calls in English that were mostly repeatable, but in Roman Urdu the same tasks were right every time for only one in eight to three in eight tasks, depending on the model and the source of variation. The failures were schema-valid errors in interpreted values, especially numbers and times written as words. Before small local agents act on Roman Urdu requests, they should be evaluated over repeated attempts and rewordings, and interpreted values should be read back to the user before the call runs.

Code and data availability

The tasks, the rewordings, all 320 raw replies, the validator pilot and the analysis code are in the folders `tool-call-consistency` and `tool-call-self-correction` of the author's research log, archived at `https://doi.org/10.5281/zenodo.24230498` and developed at `https://github.com/MHKhan50/agentic-ai-research-log`. All requests are invented; no patient data were used.

References

[1] Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ-bench: A benchmark for tool-agent-user interaction in real-world domains. arXiv:2406.12045, 2024.

[2] Muhammad Umer Tariq Butt, Stalin Varanasi, and Guenter Neumann. Roman Urdu as a low-resource language: Building the first IR dataset and baseline. In *Proceedings of the First Workshop on Advancing NLP for Low-Resource Languages*, pages 82–87, 2025. URL `https://aclanthology.org/2025.lowresnlp-1.9/`.

[3] Umer Butt, Stalin Varanasi, and Günter Neumann. Low-resource transliteration for Roman-Urdu and Urdu using transformer-based models. In *Proceedings of the Eighth Workshop on Technologies for Machine Translation of Low-Resource Languages (LoResMT 2025)*, pages 144–153, 2025. doi: 10.18653/v1/2025.loresmt-1.13.

[4] Simran Khanuja, Sandipan Dandapat, Anirudh Srinivasan, Sunayana Sitaram, and Monojit Choudhury. GLUECoS: An evaluation benchmark for code-switched NLP. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020. doi: 10.18653/v1/2020.acl-main.329.

[5] Muzammal Hussain. Roman Urdu ReAct benchmark: clinic questions in English and Roman Urdu for small local agents. Software. Zenodo, 2026.

[6] Qwen Team. Qwen2.5 technical report. arXiv:2412.15115, 2024.

[7] Aaron Grattafiori et al. The Llama 3 herd of models. arXiv:2407.21783, 2024.

[8] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, 1993.